O*NET OnLine

Network and Computer Systems Administrators

15-1244.00

Updated 2026

Install, configure, and maintain an organization's local area network (LAN), wide area network (WAN), data communications network, operating systems, and physical and virtual servers. Perform system monitoring and verify the integrity and availability of hardware, network, and server resources and systems. Review system and application logs and verify completion of scheduled jobs, including system backups. Analyze network and server resource consumption and control user access. Install and upgrade software and maintain software licenses. May assist in network modeling, analysis, planning, and coordination between network and data communications hardware and software.

Sample of reported job titles: LAN Administrator (Local Area Network Administrator), LAN Specialist (Local Area Network Specialist), Network Administrator (Network Admin), Network Analyst, Network Coordinator, Network Engineer, Network Manager, Network Specialist, Systems Engineer

Summary | Details | Custom | Easy Read | Veterans | Español

Contents

Occupation-Specific Information

Tasks

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Importance	Category	Task
89	Core	Plan, coordinate, and implement network security measures to protect data, software, and hardware.
84	Core	Perform data backups and disaster recovery operations.
78	Core	Diagnose, troubleshoot, and resolve hardware, software, or other network and system problems, and replace defective components when necessary.

Importance	Category	Task
77	Core	Maintain and administer computer networks and related computing environments, including computer hardware, systems software, applications software, and all configurations.
73	Core	Configure, monitor, and maintain email applications or virus protection software.
70	Core	Confer with network users about solutions to existing system problems.
68	Core	Operate master consoles to monitor the performance of computer systems and networks and to coordinate computer network access and use.
66	Core	Maintain logs related to network functions, as well as maintenance and repair records.
64	Core	Monitor network performance to determine whether adjustments are needed and where changes will be needed in the future.
62	Core	Design, configure, and test computer hardware, networking software and operating system software.

Occupational Requirements

Work Activities

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Importance	Work Activity
96	**Working with Computers** — Using computers and computer systems (including hardware and software) to program, write software, set up functions, enter data, or process information.
81	**Getting Information** — Observing, receiving, and otherwise obtaining information from all relevant sources.
80	**Making Decisions and Solving Problems** — Analyzing information and evaluating results to choose the best solution and solve problems.

Importance	Work Activity
78	**Analyzing Data or Information** — Identifying the underlying principles, reasons, or facts of information by breaking down information or data into separate parts.
78	**Updating and Using Relevant Knowledge** — Keeping up-to-date technically and applying new knowledge to your job.
76	**Processing Information** — Compiling, coding, categorizing, calculating, tabulating, auditing, or verifying information or data.
75	**Documenting/Recording Information** — Entering, transcribing, recording, storing, or maintaining information in written or electronic/magnetic form.
73	**Communicating with Supervisors, Peers, or Subordinates** — Providing information to supervisors, co-workers, and subordinates by telephone, in written form, e-mail, or in person.
73	**Evaluating Information to Determine Compliance with Standards** — Using relevant information and individual judgment to determine whether events or processes comply with laws, regulations, or standards.
68	**Identifying Objects, Actions, and Events** — Identifying information by categorizing, estimating, recognizing differences or similarities, and detecting changes in circumstances or events.

Detailed Work Activities

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- Implement security measures for computer or information systems.
- Develop computer or information security policies or procedures.
- Create electronic data backup to prevent loss of information.
- Resolve computer network problems.
- Resolve computer software problems.
- Troubleshoot issues with computer applications or systems.
- Maintain computer networks to enhance performance and user access.
- Provide technical support for computer network issues.
- Collaborate with others to resolve information technology issues.
- Monitor the performance of computer networks.

Work Context

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E-Mail — How frequently does your job require you to use E-mail?

100% Every day

Telephone Conversations — How often do you have telephone conversations in this job?

77% Every day

20% Once a week or more but not every day

Indoors, Environmentally Controlled — How often does this job require working indoors in an environmentally controlled environment (like a warehouse with air conditioning)?

80% Every day

Face-to-Face Discussions with Individuals and Within Teams — How frequently does your job require face-to-face discussions with individuals and within teams?

50% Every day

33% Once a week or more but not every day

13% Once a month or more but not every week

Spend Time Sitting — How much does this job require sitting?

50% Continually or almost continually

33% More than half the time

Work With or Contribute to a Work Group or Team — How important is it to work with or contribute to a work group or team in this job?

52% Extremely important

28% Very important

14% Important

Contact With Others — How much does this job require the worker to be in contact with others (face-to-face, by telephone, or otherwise) in order to perform it?

50% Constant contact with others

23% Contact with others most of the time

20% Contact with others about half the time

Determine Tasks, Priorities and Goals — How much freedom does the worker have in determining the tasks, priorities, or goals of the job?

30% A lot of freedom

50% Some freedom

13% Limited freedom

- **Freedom to Make Decisions** — How much decision making freedom, without supervision, does the job offer?
  - 30% A lot of freedom
  - 37% Some freedom
  - 30% Limited freedom
- **Importance of Being Exact or Accurate** — How important is being very exact or highly accurate in performing this job?
  - 30% Extremely important
  - 37% Very important
  - 27% Important

Experience Requirements

Job Zone

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Title	Job Zone Three: Medium Preparation Needed
Education	Most occupations in this zone require training in vocational schools, related on-the-job experience, or an associate's degree.
Related Experience	Previous work-related skill, knowledge, or experience is required for these occupations. For example, an electrician must have completed three or four years of apprenticeship or several years of vocational training, and often must have passed a licensing exam, in order to perform the job.
Job Training	Employees in these occupations usually need one or two years of training involving both on-the-job experience and informal training with experienced workers. A recognized apprenticeship program may be associated with these occupations.
Job Zone Examples	These occupations usually involve using communication and organizational skills to coordinate, supervise, manage, or train others to accomplish goals. Examples include hydroelectric production managers, desktop publishers, electricians, agricultural technicians, barbers, court reporters and simultaneous captioners, and medical assistants.
SVP Range	(6.0 to < 7.0)

Training & Credentials

State training Select a State Go

Local training ZIP Code Go

Certifications Find certifications nationwide

Apprenticeship Opportunities

All 7 displayed

Example apprenticeship titles for this occupation:

- Information Assurance Specialist
- Information Technology Specialist
- It Systems Administrator
- Network and Computer Systems Administrator (Alternate Title: It Specialist)
- Network Support Technician
- Systems Administrator
- Systems Administrator (Nof)

Specific title(s) listed above are vetted by industry and approved by the U.S. Department of Labor for use in a Registered Apprenticeship Program.

Start your career and build your skillset. Visit **Apprenticeship.gov** to learn about opportunities related to this occupation.

Worker Requirements

Software Skills

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- **Administration software** — Cisco Systems CiscoWorks; Hewlett-Packard HP Network Node Manager; Network management software; Network shutdown software
- **Configuration management software** — Chef; Patch and update management software; Perforce Helix software; Puppet
- **Data base management system software** — Amazon DynamoDB; Apache Hive; Elasticsearch; Oracle PL/SQL
- **Data base user interface and query software** — Amazon Elastic Compute Cloud EC2; Blackboard software; ServiceNow; Transact-SQL
- **Development environment software** — Apache Kafka; Apache Maven; Go; Microsoft PowerShell

- **Enterprise resource planning ERP software** — Microsoft Dynamics; Microsoft Dynamics GP; NetSuite ERP; SAP software
- **Network monitoring software** — Nagios; Quest Foglight; Remote monitoring software; Wireshark
- **Object or component oriented development software** — C#; jQuery; Perl; Swift
- **Operating system software** — Bash; Microsoft Windows Server; Red Hat Enterprise Linux; UNIX Shell
- **Web platform development software** — Apache Tomcat; Django; Microsoft ASP.NET; Node.js

Hot Technologies are requirements most frequently included across all employer job postings.
See all 95 Hot Technologies for this occupation.

In Demand skills are frequently included in employer job postings for this occupation.
See all 18 In Demand skills for this occupation.

Essential Skills

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Importance	Essential Skill
75	**Active Listening** — Giving full attention to what other people are saying, taking time to understand the points being made, asking questions as appropriate, and not interrupting at inappropriate times.
72	**Critical Thinking** — Using logic and reasoning to identify the strengths and weaknesses of alternative solutions, conclusions, or approaches to problems.
72	**Monitoring** — Monitoring/Assessing performance of yourself, other individuals, or organizations to make improvements or take corrective action.
72	**Reading Comprehension** — Understanding written sentences and paragraphs in work-related documents.
56	**Speaking** — Talking to others to convey information effectively.
53	**Active Learning** — Understanding the implications of new information for both current and future problem-solving and decision-making.
50	**Learning Strategies** — Selecting and using training/instructional methods and procedures appropriate for the situation when learning or teaching new things.
50	**Writing** — Communicating effectively in writing as appropriate for the needs of the audience.

Importance	Essential Skill
44	**Mathematics** — Using mathematics to solve problems.
28	**Science** — Using scientific rules and methods to solve problems.

Transferable Skills

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Importance	Transferable Skill
75	**Judgment and Decision Making** — Considering the relative costs and benefits of potential actions to choose the most appropriate one.
75	**Systems Analysis** — Determining how a system should work and how changes in conditions, operations, and the environment will affect outcomes.
75	**Systems Evaluation** — Identifying measures or indicators of system performance and the actions needed to improve or correct performance, relative to the goals of the system.
75	**Troubleshooting** — Determining causes of operating errors and deciding what to do about it.
72	**Complex Problem Solving** — Identifying complex problems and reviewing related information to develop and evaluate options and implement solutions.
63	**Operations Monitoring** — Watching gauges, dials, or other indicators to make sure a machine is working properly.
63	**Programming** — Writing computer programs for various purposes.
60	**Quality Control Analysis** — Conducting tests and inspections of products, services, or processes to evaluate quality or performance.
56	**Time Management** — Managing one's own time and the time of others.
53	**Equipment Maintenance** — Performing routine maintenance on equipment and determining when and what kind of maintenance is needed.

Knowledge

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Importance	Knowledge
91	**Computers and Electronics** — Knowledge of circuit boards, processors, chips, electronic equipment, and computer hardware and software, including applications and programming.
64	**Telecommunications** — Knowledge of transmission, broadcasting, switching, control, and operation of telecommunications systems.
59	**Customer and Personal Service** — Knowledge of principles and processes for providing customer and personal services. This includes customer needs assessment, meeting quality standards for services, and evaluation of customer satisfaction.
59	**English Language** — Knowledge of the structure and content of the English language including the meaning and spelling of words, and rules of composition and grammar.
54	**Administration and Management** — Knowledge of business and management principles involved in strategic planning, resource allocation, human resources modeling, leadership technique, production methods, and coordination of people and resources.
53	**Administrative** — Knowledge of administrative and office procedures and systems such as word processing, managing files and records, stenography and transcription, designing forms, and workplace terminology.
49	**Engineering and Technology** — Knowledge of the practical application of engineering science and technology. This includes applying principles, techniques, procedures, and equipment to the design and production of various goods and services.
36	**Design** — Knowledge of design techniques, tools, and principles involved in production of precision technical plans, blueprints, drawings, and models.
35	**Mathematics** — Knowledge of arithmetic, algebra, geometry, calculus, statistics, and their applications.

Importance	Knowledge
34	**Public Safety and Security** — Knowledge of relevant equipment, policies, procedures, and strategies to promote effective local, state, or national security operations for the protection of people, data, property, and institutions.

Education

How much education does a new hire need to perform a job in this occupation? Respondents said:

- **30%** Associate's degree or other 2-year degree required
- **27%** Bachelor's degree required
- **20%** Post-secondary certificate required

Worker Characteristics

Abilities

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Importance	Ability
75	**Deductive Reasoning** — The ability to apply general rules to specific problems to produce answers that make sense.
75	**Inductive Reasoning** — The ability to combine pieces of information to form general rules or conclusions (includes finding a relationship among seemingly unrelated events).
75	**Information Ordering** — The ability to arrange things or actions in a certain order or pattern according to a specific rule or set of rules (e.g., patterns of numbers, letters, words, pictures, mathematical operations).
75	**Problem Sensitivity** — The ability to tell when something is wrong or is likely to go wrong. It does not involve solving the problem, only recognizing that there is a problem.
72	**Near Vision** — The ability to see details at close range (within a few feet of the observer).
72	**Oral Comprehension** — The ability to listen to and understand information and ideas presented through spoken words and sentences.

Importance	Ability
72	**Oral Expression** — The ability to communicate information and ideas in speaking so others will understand.
72	**Written Comprehension** — The ability to read and understand information and ideas presented in writing.
56	**Fluency of Ideas** — The ability to come up with a number of ideas about a topic (the number of ideas is important, not their quality, correctness, or creativity).
53	**Flexibility of Closure** — The ability to identify or detect a known pattern (a figure, object, word, or sound) that is hidden in other distracting material.

Career Interest Types

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Occupational Interest	Career Interest Type
86	**Conventional** — Work involves following procedures and regulations to organize information or data, typically in a business setting. Conventional occupations are often associated with office work, accounting, mathematics/statistics, information technology, finance, or human resources.
53	**Realistic** — Work involves designing, building, or repairing of equipment, materials, or structures, engaging in physical activity, or working outdoors. Realistic occupations are often associated with engineering, mechanics and electronics, construction, woodworking, transportation, machine operation, agriculture, animal services, physical or manual labor, athletics, or protective services.
51	**Investigative** — Work involves studying and researching non-living objects, living organisms, disease or other forms of impairment, or human behavior. Investigative occupations are often associated with physical, life, medical, or social sciences, and can be found in the fields of humanities, mathematics/statistics, information technology, or health care service.
41	**Enterprising** — Work involves managing, negotiating, marketing, or selling, typically in a business setting, or leading or advising people in political and legal situations. Enterprising occupations are often associated with business initiatives, sales, marketing/advertising, finance, management/administration, professional advising, public speaking, politics, or law.

Occupational Interest	Career Interest Type
20	**Social** — Work involves helping, teaching, advising, assisting, or providing service to others. Social occupations are often associated with social, health care, personal service, teaching/education, or religious activities.
0	**Artistic** — Work involves creating original visual artwork, performances, written works, food, or music for a variety of media, or applying artistic principles to the design of various objects and materials. Artistic occupations are often associated with visual arts, applied arts and design, performing arts, music, creative writing, media, or culinary art.

Specific Interest Areas

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Occupational Interest	Specific Interest Area
93	**Information Technology** — Work involves designing, developing, testing, and maintaining computer software, hardware, networks, and systems.
49	**Mechanics/Electronics** — Work involves maintaining and repairing of machinery, such as automotive engines, electronic equipment, computers, and communication systems.
36	**Engineering** — Work involves applying science and technology to the design, building, testing, and use of electrical and electronic components, mechanical devices and machines, automotive, marine, and aerospace equipment and vehicles, materials, or structures.

Work Styles

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Impact	Work Style
89	**Attention to Detail** — A tendency to be detail-oriented, organized, and thorough in completing work.

Impact	Work Style
88	**Dependability** — A tendency to be reliable, responsible, and consistent in meeting work-related obligations.
78	**Cautiousness** — A tendency to be careful, deliberate, and risk-avoidant when making work-related decisions or doing work.
77	**Integrity** — A tendency to be honest and ethical at work.
73	**Intellectual Curiosity** — A tendency to seek out and acquire new work-related knowledge and obtain a deep understanding of work-related subjects.
65	**Adaptability** — A tendency to be open to and comfortable with change, new experiences, or ideas at work.
59	**Perseverance** — A tendency to exhibit determination and resolve to perform or complete tasks in the face of difficult circumstances or obstacles at work.
58	**Achievement Orientation** — A tendency to establish and maintain personally challenging work-related goals, set high work-related standards, and exert high effort toward meeting those goals and standards.
54	**Initiative** — A tendency to be proactive and take on extra responsibilities and tasks that may fall outside of one's required work role.
53	**Stress Tolerance** — A tendency to cope and function effectively in stressful situations at work.

Workforce Characteristics

Wages & Employment Trends

Median wages (2025) $47.66 hourly, $99,130 annual

State wages Select a State Go

Local wages ZIP Code Go

Employment (2025) 323,600 employees

Projected growth (2025-2035) Decline (-1% or lower)

Projected job openings 13,400

(2025-2035)

State trends Select a State Go

Top industries (2025) Professional, Scientific, and Technical Services (25% employed in this sector)
Educational Services (13%)

Source: Bureau of Labor Statistics 2025 wage data and 2025-2035 employment projections. "Projected growth" represents the estimated change in total employment over the projections period (2025-2035). "Projected job openings" represent openings due to growth and replacement.

Job Openings on the Web

State job openings Select a State Go

Local job openings ZIP Code Go

More Information

Related Occupations

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- 15-1241.00 Computer Network Architects Bright Outlook
- 15-1231.00 Computer Network Support Specialists
- 15-1211.00 Computer Systems Analysts
- 15-1299.08 Computer Systems Engineers/Architects
- 15-1232.00 Computer User Support Specialists
- 15-1242.00 Database Administrators
- 15-1243.00 Database Architects
- 15-1212.00 Information Security Analysts
- 15-1299.05 Information Security Engineers
- 15-1252.00 Software Developers

Professional Associations

Disclaimer: Sources are listed to provide additional information on related jobs, specialties, and/or industries. Links to non-DOL Internet sites are provided for your convenience and do not constitute an endorsement.

National Associations

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- American Library Association
- Association for Computing Machinery

- Healthcare Information and Management Systems Society
- IEEE Computer Society
- National Center for Women and Information Technology

Accreditation, Certification, & Unions

- CompTIA ALLY
- Building Industry Consulting Service International